

Notes: Metrick and Schmelzing (RFS, 2026)

Banking-Crisis Interventions across Time and Space

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1 Big picture

- **Question.** What kinds of banking-crisis interventions have governments used across countries and over time, what country characteristics are associated with those choices, and how large are the output losses that follow these episodes?
- **Setting.** The paper builds a new long-run database of banking-crisis interventions, covering **1,946 interventions**, **20 subcategories**, **143 countries**, and **910 crises** split into **457 canonical crises** and **453 candidate crises**.
- **Core challenge.** Existing crisis chronologies tell us when crises happened, but they do not provide a comprehensive record of *what governments actually did*, especially outside modern post-1970 samples and outside crises already recognized by the literature.
- **Main contribution.** The paper constructs a crisis-intervention database organized around a stylized financial-system balance sheet and then uses it to study intervention patterns, output losses, and long-run historical trends.
- **Main empirical result on policy mix.** Wealthier countries are more likely to use guarantees and emergency lending; guarantees are also more common in more democratic countries and in countries with more fiscal space, while lending is more common under more liberal exchange-rate regimes.
- **Main empirical result on output losses.** The paper defines a **GDP gap** as the cumulative shortfall of real GDP per capita relative to trend over the five years after a crisis. Canonical crises are associated with a GDP gap that is **17.6 percentage points** worse than the noncrisis benchmark, while candidate crises are about **9.2 percentage points** worse than noncrisis years.
- **Main historical result.** The frequency of banking interventions appears to reach an apex in the post-Bretton Woods era, but the broader historical pattern suggests a secular increase in intervention frequency since at least the late seventeenth century.
- **Why this paper matters.** It shifts attention from just dating crises to documenting the **policy menu of crisis response**. That makes it useful for work on crisis management, fiscal capacity, political economy, and long-run financial history.

2 Data and measurement (key objects)

2.1 Crisis database and unit of observation

- **Starting point: canonical crises.** The authors begin with the union of four major banking-crisis chronologies: Reinhart and Rogoff (2009), Schularick and Taylor (2012), Laeven and Valencia (2020), and Baron, Verner, and Xiong (2021). After removing overlaps, this yields **457 canonical crises**.
- **Intervention search for canonical crises.** Using the sources behind those chronologies plus an extensive primary and secondary literature, the authors identify **1,178 specific interventions** during canonical crises. In some episodes they also record deliberate nonintervention.
- **Candidate crises.** The same archival and historical sources reveal intervention episodes outside the canonical crisis dates. These are grouped into **453 candidate crises** containing **768**

interventions. The paper is explicit that these are not automatically “true” banking crises; they may be missed crises, successful preventative interventions, policy overreactions, or episodes worth further study.

- **Total database.** Combining the two sets yields **1,946 interventions** across **910 crisis episodes** in **143 countries**. The database reaches back to Roman times, though the main regression exercises are restricted to the post-1800 era.

2.2 Intervention categories and subcategories

- **Seven major categories.** The paper classifies interventions into lending, guarantees, capital injections, asset management, restructuring, rules, and other.
- **Lending subcategories.** Ad hoc emergency liquidity (AHEL), broad-based emergency liquidity (BBEL), and market liquidity assistance (MLA).
- **Guarantee subcategories.** Account guarantee (AG), asset guarantee (ASG), blanket guarantee (BG), and other liability guarantee (OLG).
- **Capital-injection subcategories.** Ad hoc capital injection (AHCI) and broad-based capital injection (BBCI).
- **Asset-management subcategories.** Ad hoc asset management (AHAM) and broad-based asset management (BBAM).
- **Restructuring subcategories.** Restructuring or resolution (RES) and stakeholder bail-in (BAIL).
- **Rules subcategories.** Debt or payments moratorium (DPM), bank suspension or bank holiday (SBH), capital and regulatory rules (CRL), and other rules (ORL).
- **Other subcategories.** Major communication (MC), stress test (ST), and other intervention (OTHER).
- **Conceptual definition.** The database includes only actions taken primarily to stabilize the financial sector. General macroeconomic policies such as fiscal stimulus, devaluation, or capital controls are intentionally excluded.

2.3 Country characteristics and covariates

- **Income.** Log real GDP per capita, mainly from long-run Maddison-style reconstructions and related historical GDP sources.
- **Debt capacity.** Gross central-government debt over GDP in the year before the crisis.
- **Government size / state capacity.** Noninterest government expenditure over GDP in the year before the crisis.
- **Political system.** The Polity score.
- **Currency regime.** A de facto exchange-rate regime measure from Ilzetzki, Reinhart, and Rogoff, where larger values correspond to more liberal or flexible regimes.

- **Candidate indicator.** A dummy that equals one for candidate crises and zero for canonical crises.
- **Year.** Crisis-year time trend used to absorb broad historical shifts.

2.4 GDP-gap measurement and long-run aggregation

- **GDP gap.** For each base year, the paper estimates trend geometric growth of real GDP per capita from the previous 15 years, extrapolates that trend five years forward, and sums the annual percentage deviations of realized GDP per capita from the extrapolated path. This cumulative five-year shortfall is the **GDP gap**.
- **Noncrisis comparison group.** The same GDP-gap calculation is repeated for noncrisis base years, which gives a benchmark distribution against which crisis years can be compared.
- **Advanced-economy bank-stress share.** For the long-run historical figure, the authors aggregate intervention observations into annual counts and compute the share of eight leading advanced economies' GDP experiencing any bank-stress episode in a given year.

3 Models and empirical specifications (all equations + interpretation)

3.1 Balance-sheet classification of interventions

The paper is not a structural macro model of crisis intervention. Its central modeling device is instead a **classification framework**: interventions are organized by where they operate on a stylized financial-system balance sheet.

- **Liabilities side.** Lending and guarantees support funding or stop runs.
- **Equity side.** Capital injections restore solvency directly.
- **Assets side.** Asset-management programs remove or restructure troubled assets.
- **Whole-balance-sheet resolution.** Restructuring and resolution reorganize failed institutions.
- **Rules and other actions.** These change the legal or regulatory environment, or attempt to stabilize beliefs through communication.

Interpretation. The point of Figure 1 is not that governments literally think in balance-sheet terms when intervening. The point is that this framework gives a coherent taxonomy for comparing interventions that otherwise look heterogeneous and historically disconnected.

3.2 Crisis-level intervention regressions

For each major intervention category g , the paper estimates a logit of the form

$$\Pr(I_{c,g} = 1 \mid X_c) = \Lambda\left(\alpha_g + \beta_{1g} \text{Candidate}_c + \beta_{2g} \log(\text{Income}_c) + \beta_{3g} \text{Polity}_c + \beta_{4g} \text{FX}_c + \beta_{5g} \frac{\text{Debt}}{\text{GDP}_c} + \beta_{6g} \frac{\text{Exp}}{\text{GDP}_c} + \beta_{7g} \text{Year}_c\right).$$

where $I_{c,g}$ equals one if crisis c uses intervention category g and zero otherwise.

- **Unit of observation.** A crisis episode.
- **Dependent variables.** Separate logits for guarantees, lending, capital injections, restructuring, asset management, rules, and other.
- **Timing.** The regressors are measured in $t - 1$, the year before the crisis begins.
- **Interpretation.** These are reduced-form correlations about the kinds of countries that use certain crisis tools more often. They are not causal estimates of the effect of country characteristics on intervention choice.

3.3 GDP-gap construction

Let y_t denote real GDP per capita in the base year t . Let $\hat{g}_t$ be the estimated trend geometric growth rate from the previous 15 years. The paper projects a trend path

$$\hat{y}_{t+h|t} = y_t e^{h\hat{g}_t}, \quad h = 0, 1, 2, 3, 4,$$

and defines the GDP gap as the cumulative deviation from trend over the subsequent five years:

$$\text{GDPGap}_t = \sum_{h=0}^4 \frac{y_{t+h} - \hat{y}_{t+h|t}}{y_t}.$$

- **Interpretation.** This is the “area under the trend GDP line.” More negative values mean larger medium-run output losses following the crisis.
- **Why this measure is useful.** It summarizes the depth and persistence of postcrisis weakness in one number rather than focusing only on the immediate output drop.

3.4 GDP-gap regressions

The paper then relates GDP losses to the same crisis and country characteristics through regressions of the form

$$\begin{aligned} \text{GDPGap}_c = & \alpha + \delta \text{Candidate}_c + \gamma_1 \log(\text{Income}_c) + \gamma_2 \text{Polity}_c + \gamma_3 \text{FX}_c \\ & + \gamma_4 \frac{\text{Debt}}{\text{GDP}_c} + \gamma_5 \frac{\text{Exp}}{\text{GDP}_c} + \gamma_6 \text{Year}_c + u_c. \end{aligned}$$

with versions estimated on the combined sample and on canonical/candidate subsamples.

- **Interpretation of signs.** Since GDP gaps are typically negative in crises, a *negative* coefficient implies more severe losses, while a positive coefficient implies smaller losses.
- **Main use.** These regressions ask whether the severity of crisis aftermaths systematically varies with a country’s wealth, politics, exchange-rate regime, or fiscal position.

3.5 Long-run historical aggregation

For the long-run time series, the authors aggregate intervention activity at the annual level. Let N_t be the total number of interventions in year t , and let $\omega_{i,t}$ denote country i ’s GDP share among the selected advanced economies. Then an annual advanced-economy bank-stress share can be summarized as

$$B_t = \sum_{i \in \mathcal{D}} \omega_{i,t} \mathbf{1}\{\text{country } i \text{ has any crisis intervention in year } t\},$$

where $\mathcal{D}$ is the set of eight leading advanced economies used in the historical exercise. The figure reports both N_t and an 11-year centered moving average of B_t .

- **Interpretation.** The count series measures the absolute intensity of interventions, while the GDP-weighted series measures how much of the advanced world was under bank stress in a given year.

4 Identification and instruments (what makes the estimates credible)

4.1 This is a descriptive-identification paper, not an IV paper

- **No instruments.** The paper does not claim causal identification through an instrument or a natural experiment.
- **What it identifies.** Its main contribution is measurement and classification: a systematic intervention database, a clear separation between canonical and candidate crises, and reduced-form patterns linking interventions and crisis severity to country characteristics.

4.2 What makes the evidence informative

- **Consistent coding of interventions.** The seven-category / 20-subcategory classification is applied across centuries and countries.
- **Canonical versus candidate split.** This is important because it allows the authors to ask whether interventions outside the standard crisis chronologies look economically meaningful.
- **Precrisis covariates.** Country regressors are measured in the year before the crisis, which helps separate policy choice correlates from immediate within-crisis outcomes.
- **Noncrisis benchmark for GDP losses.** The GDP-gap framework gives a useful benchmark against ordinary years, not just comparisons among different crises.
- **Long historical perspective.** Because the database extends far back in time, the paper can place the global financial crisis and post-1980 experience in a much longer-run context.

4.3 Main caveats

- **Candidate crises are ambiguous by construction.** Some may be true missed crises, some may be successful preventive interventions, and some may be policy overreactions or unrelated rent extraction.
- **Observations are not independent.** The authors explicitly warn that crisis-level regressions involve repeated countries and clustered global events, so standard errors should be interpreted cautiously.
- **Intervention taxonomy is necessarily partial.** The database excludes broad macro policies, so it is not a complete accounting of all crisis responses.
- **Historical measurement limits.** Earlier centuries have sparser records, especially for output, debt, and government-spending variables.

4.4 Relation to the literature

- The paper complements, rather than replaces, crisis chronologies such as Reinhart–Rogoff, Schularick–Taylor, Laeven–Valencia, and Baron–Verner–Xiong.
- Relative to Laeven and Valencia in particular, the paper’s novelty is the deeper classification of interventions and the inclusion of interventions outside standard crisis windows.
- Conceptually, it is close to the literature on crisis containment and resolution, but its distinctive contribution is to move from event dating to **policy-action mapping**.

5 Tables and figures

5.1 Figure 1: Overview of the major intervention categories and subcategories

Read:

- This is the paper’s conceptual map. It places each intervention category on a stylized financial-system balance sheet.
- The figure makes clear that modern crisis management is much broader than a lender-of-last-resort story. Governments intervene on assets, liabilities, equity, regulation, and even expectations.
- The 20 subcategories matter because the same major category can take very different forms: for example, lending can be ad hoc, broad based, or aimed at market liquidity assistance.

5.2 Table 1: Intervention types by period and category

Read:

- Over the full sample, lending is the largest category at about **28.9%** of all interventions, followed by guarantees (**16.3%**), capital injections (**14.8%**), rules (**14.4%**), restructuring (**13.0%**), asset management (**4.2%**), and other (**2.7%**).

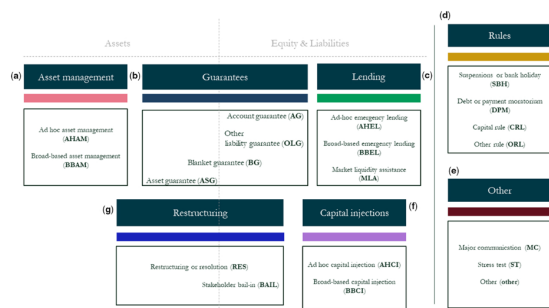


Figure 1
Overview of the major intervention categories and subcategories (the balance sheet view)
The figure displays an organigram featuring the seven major banking intervention categories, including (clockwise from left to right) asset management (labelled “a”), guarantees (b), lending (c), rules (d), other (e), capital injections (f), restructuring (g). Each of these categories is overlaying the area of the balance sheet that the category targets (assets, equity and liabilities, and other) and the category boxes featuring the 20 subcategories defined further in the text. *Source: Authors’ creation.*

Table 1
Intervention types by period and category
Intervention distribution by period

Share of all interventions (n)	Pre1800	1800–1869	1870–1913	1914–1945	1946–1971	1972–1999	2000–2019	Total
Lending	2.9 (57)	4.3 (84)	4.2 (82)	6.3 (122)	0.1 (14)	6.1 (118)	4.4 (85)	28.9 (562)
AHEL	1.8 (40)	2.6 (50)	1.9 (36)	3.4 (64)	0.0 (2)	2.6 (53)	1.5 (29)	13.8 (274)
BBEL	0.7 (14)	1.4 (27)	2.0 (39)	2.8 (54)	0.1 (10)	3.3 (62)	2.3 (42)	13.0 (248)
MLA	0.1 (3)	0.3 (7)	0.4 (7)	0.2 (4)	0.0 (2)	0.1 (3)	0.8 (14)	1.9 (40)
Capital injections	0.3 (6)	0.3 (5)	0.5 (9)	2.2 (37)	0.1 (10)	6.2 (120)	5.4 (101)	14.8 (288)
AHCI	0.3 (6)	0.3 (5)	0.4 (7)	1.6 (29)	0.0 (3)	3.4 (68)	3.4 (64)	9.4 (182)
BBCI	0.0 (0)	0.0 (0)	0.1 (2)	0.5 (8)	0.1 (7)	2.8 (52)	2.0 (37)	5.6 (106)
Guarantees	1.4 (27)	1.0 (16)	1.7 (30)	2.7 (52)	0.0 (2)	3.7 (89)	5.2 (101)	16.3 (317)
AG	1.0 (17)	0.2 (4)	0.7 (13)	1.0 (19)	0.0 (2)	2.2 (46)	1.6 (32)	6.9 (133)
ASG	0.1 (1)	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)	1.0 (12)	0.8 (16)	1.5 (29)
BG	0.1 (1)	0.1 (2)	0.2 (4)	0.3 (5)	0.0 (0)	1.0 (13)	0.2 (5)	1.5 (30)
OLG	0.5 (8)	0.6 (10)	0.7 (13)	1.6 (28)	0.0 (0)	1.0 (18)	2.4 (48)	6.9 (125)
Rules	2.6 (55)	2.8 (50)	2.0 (39)	3.3 (64)	0.1 (6)	2.2 (41)	1.1 (26)	14.4 (281)
DPM	1.0 (22)	0.2 (5)	0.3 (5)	0.8 (17)	0.0 (1)	0.3 (6)	0.0 (0)	2.6 (56)
SBH	1.2 (22)	1.5 (28)	0.8 (16)	1.7 (35)	0.1 (3)	0.8 (14)	0.3 (5)	6.5 (123)
CRL	0.1 (1)	0.2 (3)	0.0 (2)	0.0 (1)	0.0 (0)	0.1 (4)	0.6 (13)	1.1 (24)
ORL	0.5 (10)	0.7 (14)	0.8 (16)	0.6 (11)	0.0 (2)	1.0 (17)	0.3 (8)	4.0 (78)
Restructuring	1.3 (25)	1.3 (26)	0.7 (12)	1.6 (31)	0.1 (6)	5.3 (111)	2.1 (41)	13.0 (252)
RES	0.6 (14)	0.8 (17)	0.6 (11)	1.5 (30)	0.1 (6)	4.9 (101)	1.1 (22)	10.3 (201)
BAIL	0.5 (11)	0.5 (9)	0.0 (1)	0.0 (0)	0.0 (0)	0.5 (10)	1.0 (19)	2.5 (51)
Asset Management	0.0 (2)	0.0 (0)	0.2 (3)	0.2 (13)	0.0 (0)	2.2 (39)	1.3 (25)	4.2 (82)
AHAM	0.0 (0)	0.0 (0)	0.0 (0)	0.1 (2)	0.0 (0)	0.2 (5)	0.5 (8)	0.8 (15)
BBAM	0.0 (2)	0.0 (0)	0.2 (3)	0.1 (11)	0.0 (0)	2.0 (34)	0.9 (17)	3.5 (67)
Other	0.6 (10)	0.1 (4)	0.5 (9)	0.3 (8)	0.0 (2)	0.5 (11)	0.2 (8)	2.7 (52)
MC	0.0 (1)	0.0 (0)	0.0 (0)	0.0 (1)	0.0 (1)	0.0 (2)	0.1 (2)	0.2 (7)
ST	0.0 (0)	0.1 (1)	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (2)	0.0 (1)	0.2 (4)
OTHER	0.5 (9)	0.1 (3)	0.5 (9)	0.3 (7)	0.0 (1)	0.4 (7)	0.1 (5)	2.1 (41)
NOI	0.2 (4)	0.7 (13)	1.5 (28)	1.0 (19)	0.0 (1)	2.4 (45)	0.1 (2)	5.8 (112)
Total	9.6 (186)	10.2 (199)	11.0 (215)	17.7 (344)	2.1 (40)	29.4 (572)	20.0 (390)	100.0 (1946)

Figures in brackets measure the absolute number of interventions, and those outside of brackets indicate the percentage of all-time total interventions. The intervention tags correspond to the subgroups (see Figure 1 and discussion in the introduction): AG: account guarantee, AHAM: ad hoc asset management, AHCI: ad hoc capital injection, AHEL: ad hoc emergency liquidity, ASG: asset guarantee, BAIL: stakeholder bail-in, BBAM: broad-based asset management, BBAI: broad-based capital injection, BBEL: broad-based emergency liquidity, BG: blanket guarantee, CRL: capital and regulatory rules, DPM: debt or payments moratorium, MC: major communication, MLA: market liquidity assistance, NOI: no intervention (not included in total), OLG: other liability guarantee, ORL: other rules, OTHER: other intervention, RES: restructuring or resolution, SBH: bank suspension or bank holiday, and ST: stress test.

- Before 1945, the intervention mix is dominated by rules and lending. Together they make up more than half of all interventions in each early subperiod.
- After 1945, rules and lending become less dominant, while capital injections rise sharply. Since 1972, capital injections are the largest category, and since 2000 they account for more than a quarter of all interventions.
- Guarantees become more important again over the past two decades, regaining a prominence not seen since the pre-1750 period.

5.3 Figure 2: Intervention policy mix by crisis type and by economic, fiscal, and political variables

Read:

- The intervention mix looks broadly similar between canonical and candidate crises, though canonical crises rely relatively more on **broad-based emergency lending**, while candidate crises lean more toward **ad hoc emergency lending**.
- High-income countries rely much more on **capital injections and guarantees**; in low-income countries those two categories account for less than a quarter of interventions.
- In lower-income countries, the dominant crisis tools are **rules and lending**, which together make up roughly two-thirds of interventions.
- Fixed exchange-rate regimes appear less able to rely on emergency lending than liberal exchange-rate regimes, foreshadowing the regression results.

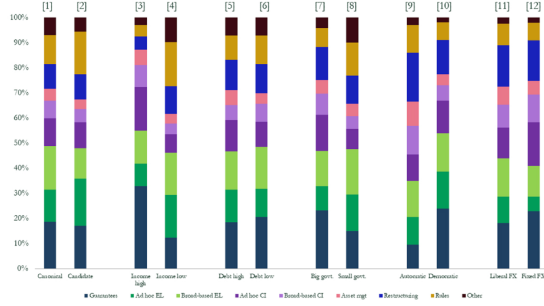


Figure 2
Intervention policy mix by crisis type and by economic, fiscal, and political variables, 33 AD–2019 BC
The color code in this Figure follows the color code introduced for the nine broad intervention categories in Figure 1, with Guarantees (dark blue), Ad hoc emergency lending (dark green), broad-based emergency lending (light green), Ad hoc capital injections (dark purple), broad-based capital injections (light purple), Asset management (pink), Restructuring (light blue), Rules (yellow), and Other (dark red). For full notes on the historical real GDP data used, refer to details in the text and country notes in Internet Appendix F. Overwhelmingly, the real per capita series on the basis of the latest Maddison Project estimates are used as well as the underlying sources therein: the real per capita levels are benchmarked to 2011 US\$ levels. Readers should consult the full documentation and notes in Bolt and van Zanden (2020) and associated online spreadsheets via Maddison Project 2020 release (<https://www.rug.nl/ggdc/historicaldevelopment/maddison/releases/maddison-project-database-2020>). We use Nuvolari and Ridolfi's (2021) GDP data for France and Pfister's (2022) data for Germany, both of which are benchmarked to 1990 Geary–Khamis Dollar (\$GK) levels. The real per-capita sample has a different source basis from our GDP sample detailed in Internet Appendix E, with details on historical GDP bases and calculations there. The interventions without income data points are excluded. The FX regime "Liberal FX" denotes currency regime observations on the "fine classification" basis in Ilzetzki, Reinhart, and Rogoff (2019) with values ≥ 7 ; "Fixed FX" with values < 7 . For the Big govt. and Small govt. as well as the Debt high and Debt low samples, the samples are split at the median value of all of the observations. The no intervention observations are included in the "other" category throughout. Source: Authors' creation.

Intervention type	Guarantees	Lending	Capital injection	Restructuring	Asset mgt.	Rules	Other
Candidate	-1.383*** (0.302)	-0.684** (0.270)	-0.918*** (0.281)	-0.176 (0.272)	-1.664*** (0.435)	0.610 (0.385)	-0.052 (0.574)
Income (log)	0.613*** (0.226)	0.643*** (0.201)	0.258 (0.196)	-0.447** (0.200)	0.404 (0.246)	-0.016 (0.262)	-0.148 (0.383)
Polity	0.100*** (0.033)	0.040 (0.027)	-0.041 (0.027)	0.036 (0.029)	-0.063* (0.033)	0.064 (0.043)	0.072 (0.067)
FX regime	-0.015 (0.034)	0.089*** (0.031)	-0.057* (0.031)	-0.011 (0.031)	0.028 (0.041)	0.046 (0.044)	0.121* (0.070)
Debt/GDP	-0.014*** (0.005)	-0.003 (0.004)	0.0001 (0.004)	-0.007* (0.004)	0.002 (0.005)	0.007 (0.006)	0.008 (0.008)
Exp/GDP	-0.004 (0.016)	-0.023 (0.015)	0.022 (0.015)	0.032** (0.016)	-0.026 (0.020)	-0.045** (0.023)	-0.033 (0.032)
Year	0.015 (0.012)	-0.009 (0.010)	0.021* (0.011)	-0.009 (0.010)	0.043** (0.019)	0.015 (0.015)	-0.001 (0.022)
Constant	-34.7 (23.0)	12.3 (20.1)	-43.5** (22.1)	21.0 (20.5)	-88.9 (37.8)	-31.3 (29.0)	-0.82 (43.6)
N	272	272	272	272	272	272	272

Table 2 displays the regression results at the crisis level, for which each crisis (candidate or canonical) is one observation. Data begin in 1800. Each column represents a single regression, with the dependent variable given in the top row. The intervention categories in the first row follow the category definitions given in the introduction. For each regression, the dependent variable is equal to one if that category of intervention is present in a crisis and zero otherwise. *Income* is measured in logs, using data from the same year as the dependent variable. *Candidate* takes a value of one for a candidate crisis and zero for a canonical crisis. Sources for the income data are discussed in Internet Appendix F, and robustness checks are found in Internet Appendix C. *Polity* represents numerically the polity score per data in Marshall and Gurr (2020). *Liberal FX regime* represents the currency regime variable, sourced from Ilzetzki, Reinhart, and Rogoff (2019). *Debt/GDP* represents total gross central government debt / GDP (domestic + external), sourced from Reinhart and Rogoff (2009). *Exp/GDP* represents noninterest government expenditures / GDP, sourced from Mauro et al. (2015). We exclude all observations for which at least one of the variables in *Income*, *Debt/GDP*, or *Exp/GDP* is missing. All of the variables are measured for $t-1$, the year preceding the crisis start date. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.4 Table 2: Regression results for the major intervention categories

Read:

- The strongest and clearest regression results are for **guarantees** and **lending**.
- Guarantees are more likely in richer countries, in more democratic countries, and in countries with lower debt-to-GDP. Candidate crises are less likely than canonical crises to involve guarantees.
- Lending is more likely in richer countries and in countries with more liberal exchange-rate regimes.
- The table should be read as reduced-form correlation evidence, not as a causal intervention-choice model.

5.5 Figure 3: GDP gaps associated with financial crises, subtypes, and noncrisis years

Read:

- All crisis years are associated with negative GDP gaps on average, while noncrisis base years have a positive mean GDP gap because trend growth is positive over the sample.
- Canonical crises are much more severe: mean GDP gap about **-12.7%**, median about **-11.1%**.
- Candidate crises are milder but still economically meaningful: mean GDP gap about **-4.3%**, median about **-5.1%**.
- The key message is that candidate crises are not observational noise in aggregate: they sit roughly halfway between canonical crises and noncrisis years.

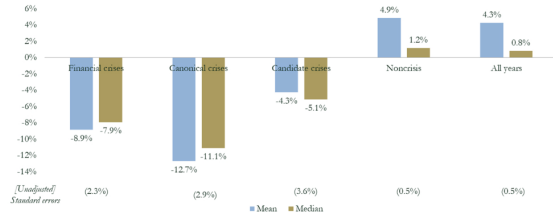


Figure 3
GDP gaps associated with financial crises, subtypes, and noncrisis years
This figure displays the mean and median GDP gap sums associated with different categories of country-years, including those associated with all financial crises (columns 1–2), only canonical crises (3–4), only candidate crises (5–6), all noncrisis years (7–8) and all years (9–10). “GDP gap sum” is defined as the sum of annual real GDP per capita difference to GDP growth trend-implied real per capita GDP, over t to $t+4$, with the trend path using the realized (log) real GDP path over $t-14$ to t where t = the crisis start year. Income data as per sources in Internet Appendix F. The Financial crisis sample includes all candidate and canonical crises. All years includes all of the annual-level GDP gap sum observations for which underlying (log) real per capita GDP is available for the countries in the intervention sample via Bolt and van Zanden (2020) and sources therein. Source: Authors’ creation.

Table 3
Regression results for all crises, by crisis type, and dependent variables

	Combined	Canonical crises		Candidate crises	
Candidate	0.1464 (0.0755)	0.0685 (0.0596)	—	—	—
Income (log)	-0.2463*** (0.0379)	-0.2733*** (0.0426)	-0.2306*** (0.0416)	-0.2815*** (0.0585)	-0.3621*** (0.0791)
Polity	0.0071 (0.0055)	0.0105* (0.0060)	0.0160 (0.0074)	0.0133 (0.0083)	-0.0028 (0.0087)
FX regime	0.0114 (0.0078)	0.0179** (0.0068)	0.0123 (0.0082)	0.0239*** (0.0092)	0.0108 (0.0156)
Debt/GDP		-0.0012 (0.0009)		-0.0018 (0.0011)	0.0005 (0.0017)
Exp/GDP		0.0054 (0.0034)		0.0061 (0.0042)	0.0046 (0.0058)
Year	0.0022 (0.0028)	0.0029 (0.0023)	0.0033 (0.0043)	0.0031 (0.0043)	0.0049 (0.0041)
Constant	-3.922 (5.4536)	-3.7137 (4.5573)	-4.7344 (8.4863)	-4.0594 (8.5654)	-6.5296 (8.0010)
N	334	259	203	146	131

Table 3 displays the regression results at the crisis level, for which each crisis (candidate or canonical) is one observation. Data begin in 1800. Each column represents a single regression, with the dependent variable given in the top row. The crises types in the first row follow the crises definitions given in the introduction. For each regression, the dependent variable is the GDP gap as defined in the associated Section 3 text. *Income* is measured in logs, using data from the same year as the dependent variable. *Candidate* takes on a value of one for a candidate crisis, and zero for a canonical crisis. Sources for the income data are discussed in Internet Appendix F. *Polity* represents the numerical polity score per data in Marshall and Gurr (2020). *Liberal FX regime* represents the currency regime variable, sourced from Ilzetzki, Reinhart, and Rogoff (2019). *Debt/GDP* represents total gross central government debt / GDP (domestic + external), sourced from Reinhart and Rogoff (2009). *Exp/GDP* represents noninterest government expenditures / GDP, sourced from Mauro et al. (2015). We exclude all observations for which at least one of the variables *Income*, *Debt/GDP*, or *Exp/GDP* is missing. All of the variables are measured for $t-1$, the year preceding the crisis start date. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.6 Table 3: Regression results for GDP gaps

Read:

- Higher-income countries experience **more negative GDP gaps**; the income coefficient is negative and significant across the main specifications.
- There is some evidence, especially for canonical crises, that more democratic political systems and more liberal exchange-rate regimes are associated with **smaller output losses**.
- The candidate-crisis indicator is positive in the combined sample, consistent with candidate crises being less severe than canonical crises even conditional on observables.
- These results are economically suggestive, but the authors caution against overinterpreting formal statistical significance because the crisis observations are not independent.

5.7 Figure 4: Distribution of total interventions by year and by share of advanced-economy GDP exhibiting bank stress

Read:

- This is the long-run historical figure of the paper. It shows both the annual number of interventions and the GDP-weighted share of advanced economies under bank stress from **1665 to 2019**.
- The post-Bretton Woods era is an obvious apex, but the broader pattern is a secular increase over the long run rather than a purely modern anomaly.
- The paper highlights that only about **1.7%** of advanced-economy GDP was under bank stress on average before 1701, versus about **6.1%** in the first half of the eighteenth century.
- The figure also reveals historically underappreciated peaks, such as 1810–1821 and 1847–1848, not only the twentieth-century episodes and the global financial crisis.

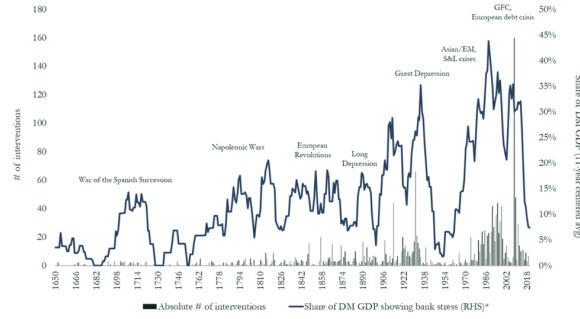


Figure 4
Distribution of total interventions by year and by share of advanced economy GDP exhibiting “bank stress,” 1665–2019
The left axis in the figure provides an intervention-level aggregation and reports the total number of interventions in each year (t /# of annual events”) for the full database, summing all canonical and candidate crises intervention observations for each year. The right axis provides a GDP-weighted view for advanced economies, focused on the subset of eight leading DM countries (Italy, the UK, Netherlands, France, Germany, Spain, the US [from 1790], and Japan [from 1870]). Canonical or candidate crises intervention observations in a year for one of the eight countries results in a positive observation for the year of its respective DM GDP share, with the shares for all eight countries then summed for each year. The 11-year centered moving average is displayed for this measure. GDP weights based on Schmelzing (2020). Source: Authors’ creation.

6 Short summary

- **Conclusion.** The paper creates a comprehensive database of banking-crisis interventions and uses it to document how crisis responses vary across countries, institutions, and time.
- **Main identification insight.** The contribution is not an instrument-based causal design but a careful measurement framework: classify interventions systematically, separate canonical from candidate crises, and compare intervention choices and output losses across crisis episodes using precrisis country characteristics.
- **Main empirical lesson.** Crisis management is historically diverse, not just lender-of-last-resort lending. Wealth, politics, fiscal space, and exchange-rate regime shape the intervention menu, and output losses are economically large even for many candidate crises.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that one can systematically map the policy responses to banking distress across centuries and countries using a coherent intervention taxonomy.
- It shows that crisis response is not uniform: richer and more democratic countries with more room in the fiscal and monetary regime space tend to use a different menu of interventions.
- It also shows that crisis episodes outside the standard chronologies are economically meaningful enough to warrant serious study.

7.2 Economic intuition

- Different intervention tools require different state capacities. Guarantees require fiscal credibility; emergency lending is harder under fixed exchange-rate regimes; capital injections become more feasible in richer financial systems with deeper fiscal and institutional capacity.

- Candidate crises matter because banking distress can be partly hidden by the success of the intervention itself. If policy prevented the full crystallization of panic or output collapse, the episode may never enter a standard crisis chronology.
- The historical trend toward more interventions likely reflects both growing financial complexity and the expansion of the state toolkit for dealing with financial instability.

7.3 Takeaway

This paper is best read as a long-run empirical map of how governments actually fight banking crises. Its major innovation is to move beyond crisis dating toward intervention coding. That shift matters because the policy menu itself contains information: it tells us which tools countries can credibly deploy, how state capacity shapes crisis management, and why some historically important episodes may be missing from standard crisis lists. For work on financial crises, bailout design, guarantees, bank resolution, fiscal capacity, or macro-financial history, this paper provides a new benchmark dataset and a useful organizing framework.